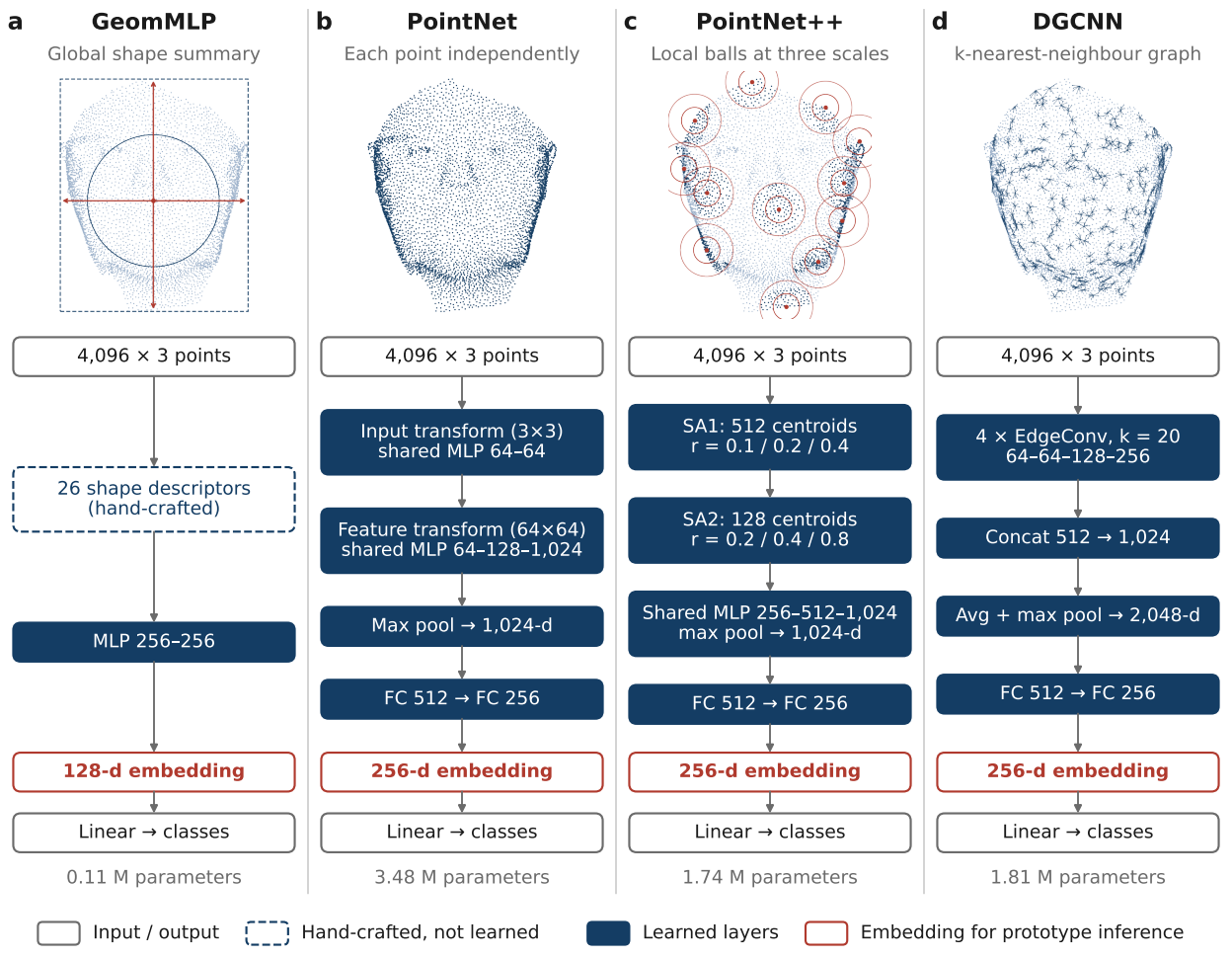

a
GeomMLP
Global shape summary
4,096 × 3 points
26 shape descriptors (hand-crafted)
MLP 256–256
128-d embedding
Linear → classes
0.11 M parameters
b
PointNet
Each point independently
4,096 × 3 points
Input transform (3×3) shared MLP 64–64
Feature transform (64×64) shared MLP 64–128–1,024
Max pool → 1,024-d
FC 512 → FC 256
256-d embedding
Linear → classes
3.48 M parameters
c
PointNet++
Local balls at three scales
4,096 × 3 points
SA1: 512 centroids r = 0.1 / 0.2 / 0.4
SA2: 128 centroids r = 0.2 / 0.4 / 0.8
Shared MLP 256–512–1,024 max pool → 1,024-d
FC 512 → FC 256
256-d embedding
Linear → classes
1.74 M parameters
d
DGCNN
k-nearest-neighbour graph
4,096 × 3 points
4 × EdgeConv, k = 20 64–64–128–256
Concat 512 → 1,024
Avg + max pool → 2,048-d
FC 512 → FC 256
256-d embedding
Linear → classes
1.81 M parameters
Input / output
Hand-crafted, not learned
Learned layers
Embedding for prototype inference